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[22EC34]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

DIGITAL SYSTEM DESIGN USING VERILOG

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 X 2=10**		
1.	**a.**	Differentiate between loosely typed and strongly typed languages?	L2	CO1	**02**
	b.	Draw the gate level schematic of a full adder using gate primitives supported by Verilog.	L3	CO1	**02**
	c.	Describe how is an always block different from an initial block.	L2	CO2	**02**
	d.	Write a Verilog code to implement a 2:1 MUX using ternary operator.	L3	CO5	**02**
	e.	Explain the functionality of a task in Verilog.	L2	CO6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the following with an example. **i.** Text Macros **ii.** Number format and its representation.	L2	CO1	**05**
		OR			
	b.	With schematic and truth table, write a Verilog program to implement a 4:1 MUX.	L3	CO1	**05**
3.	**a.**	What is a race condition in Verilog? Explain using multiple initial blocks as an example.	L2	CO2	**05**
		OR			
	b.	Explain the functionality of a begin-end block with an example.	L2	CO2	**05**
4.	**a.**	Outline the various display tasks supported by Verilog with an example.	L2	CO3	**05**
		OR			
	b.	With an example shows how to write to data into multiple files and also print on the screen.	L2	CO3	**05**
				CO4	
5.	**a.**	Outline the various types of nets available in Verilog.	L2		**05**
		OR			
	b.	Explain the differences between blocking and non-blocking assignment in Verilog with suitable examples.	L2	CO4	**05**
6.	**a.**	Outline various bitwise logical and arithmetic operators supported by Verilog.	L2	CO5	**05**

		OR			
	b.	Discuss various logical reduction operators and equality operators in Verilog.	L3	CO5	**05**
7.	**a.**	Describe how functions are different from tasks.	L2	CO6	**05**
		OR			
	b.	Illustrate the usage of case statement with syntax with an example of 4:1 Mux.	L2	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	With truth table and gate level schematic write a Verilog code for a 1-bit full adder. Using the 1-bit full adder, construct a 4-bit full adder structural model in Verilog with a neat diagram.	L3	CO1	**10**
		OR			
	b.	Write a Verilog code in gate level to implement 2:1 MUX with the schematic and structural model of 4:1 MUX using 2:1 MUX along with the block diagram.	L3	CO1	**10**
9.	**a.**	Write a Verilog behavioral code to implement an 8:3 encoder along with truth table and a test bench for it.	L3	CO2	**10**
		OR			
	b.	Write a Verilog behavioral code to implement an 8:1 MUX along with truth table and a test bench for it.	L3	CO2	**10**
10.	**a.**	Explain the various screen and file output commands with an example for each one of them.	L2	CO3	**10**
		OR			
	b.	Explain each of the format specifiers used in Verilog	L2	CO3	**10**
11.	**a.**	Describe the usage of parameters and events in Verilog with an example.	L3	CO4	**10**
		OR			
	b.	Explain the following data types with an example. **i.** reg **ii.** integer **iii.** real **iv.** time **v.** string	L2	CO4	**10**
12.	**a.**	Solve the following expressions and write the output if	L3	CO5	**10**

		A=4, B= 3, X=4'b1010, Y= 4'b1101 and Z= 4'b10x1 **i.** X & Y **ii.** X | Y **iii.** A == B **iv.** A!=B **v.** &X **vi.** &Y **vii.** X==Y **viii.** X!=Y **ix.** |Z **x.** X+Y			
		OR			
	b.	**i.** Write a Verilog code to perform 128-Bit addition using continuous assignment. **ii.** Illustrate various arithmetic operators in Verilog with a program and output.	L3	CO5	**10**
13.	**a.**	Explain various loop constructs supported by Verilog with an example for each one of them.	L3	CO6	**10**
		OR			
	b.	**i.** Write a Verilog task to add two numbers and display the output. **ii.** Write a Verilog code implement Mux with Function and Continuous Assignment.	L3	CO6	**10**